COMPUTING THE PULLBACK OF A SHEAF ALONG A MORPHISM FROM A COLIMIT OF SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The construction of the functors to the colimit site. Same chapter, Lemma lemma-colimit-sites. Supplies the functors u_i and the continuity checked at the start of the proof.
- Exactness of adjoints. Categories, Lemma categories-lemma-exact-adjoint. Shows that the pullback, being a left adjoint, is right exact, so that only left exactness has to be proved.
- Finite limits of sheaves and commutation of filtered colimits with finite limits. Same chapter, Lemma lemma-limit-sheaf, and Categories, Lemma categories-lemma-directed-commutes. Reduce left exactness to the displayed formula, in which filtered colimits commute with finite limits.

Source reference. Original lemma 09YL: sources/stacks-sites/pinned-original-source.tex, source lines 4079–4079. Navigation pointer invoked at proof line 4079; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0039: sources/stacks-sites/pinned-original-source.tex, source lines 4096–4096. Navigation pointer invoked at proof line 4096; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 002W: sources/stacks-sites/pinned-original-source.tex, source lines 4054–4054. Navigation pointer invoked at proof line 4054; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 4174–4174. Navigation pointer invoked at proof line 4174; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Situation 0.1. Here we are given

(1) a cofiltered index category $\mathcal{I}$,
(2) for $i \in \mathrm{Ob}(\mathcal{I})$ a site $\mathcal{C}_i$ such that every covering in $\mathcal{C}_i$ has a finite index set,
(3) for a morphism $a : i \to j$ in $\mathcal{I}$ a morphism of sites $f_a : \mathcal{C}_i \to \mathcal{C}_j$ given by a continuous functor $u_a : \mathcal{C}_j \to \mathcal{C}_i$,

such that $f_a \circ f_b = f_c$ whenever $c = a \circ b$ in $\mathcal{I}$.

Lemma 0.2. *In Situation 0.1 let $u_i : \mathcal{C}_i \to \mathcal{C}$ be as constructed in Lemma 09YL. Then u_i defines a morphism of sites $f_i : \mathcal{C} \to \mathcal{C}_i$. For $U_i \in \mathrm{Ob}(\mathcal{C}_i)$ and sheaf $\mathcal{F}$ on $\mathcal{C}_i$ we have*

$$f_i^{-1}\mathcal{F}(u_i(U_i)) = \mathrm{colim}_{a:j\to i} f_a^{-1}\mathcal{F}(u_a(U_i)) \tag{0.2.1}$$

Proof. It is immediate from the arguments in the proof of Lemma 09YL that the functors u_i are continuous. To finish the proof we have to show that $f_i^{-1} := u_{i,s}$ is an exact functor $Sh(\mathcal{C}_i) \to Sh(\mathcal{C})$. In fact it suffices to show that f_i^{-1} is left exact, because it is right exact as a left adjoint (Categories, Lemma 0039). We first prove (0.2.1) and then we deduce exactness.

For an arbitrary object V of $\mathcal{C}$ we can pick a $a : j \to i$ and an object $V_j \in \mathrm{Ob}(\mathcal{C})$ with $V = u_j(V_j)$. Then we can set

$$\mathcal{G}(V) = \mathrm{colim}_{b:k\to j} f_{a\circ b}^{-1}\mathcal{F}(u_b(V_j))$$

The value $\mathcal{G}(V)$ of the colimit is independent of the choice of $b : j \to i$ and of the object V_j with $u_j(V_j) = V$; we omit the verification. Moreover, if $\alpha : V \to V'$ is a morphism of $\mathcal{C}$, then we can choose $b : j \to i$ and a morphism $\alpha_j : V_j \to V_j'$ with $u_j(\alpha_j) = \alpha$. This induces a map $\mathcal{G}(V') \to \mathcal{G}(V)$ by using the restrictions along the morphisms $u_b(\alpha_j) : u_b(V_j) \to u_b(V_j')$. A check shows that $\mathcal{G}$ is a presheaf (omitted). In fact, $\mathcal{G}$ satisfies the sheaf condition. Namely, any covering $\mathcal{U} = \{U^t \to U\}$ in $\mathcal{C}$ comes from a finite level. Say $\mathcal{U}_j = \{U_j^t \to U_j\}$ is mapped to $\mathcal{U}$ by u_j for some $a : j \to i$ in $\mathcal{I}$. Then we have

$$H^0(\mathcal{U}, \mathcal{G}) = \mathrm{colim}_{b:k\to j} H^0(u_b(\mathcal{U}_j), f_{b\circ a}^{-1}\mathcal{F}) = \mathrm{colim}_{b:k\to j} f_{b\circ a}^{-1}\mathcal{F}(u_b(U_j)) = \mathcal{G}(U)$$

as desired. The first equality holds because filtered colimits commute with finite limits (Categories, Lemma 002W). By construction $\mathcal{G}(U)$ is given by the right hand side of (0.2.1). Hence (0.2.1) is true if we can show that $\mathcal{G}$ is equal to $f_i^{-1}\mathcal{F}$.

In this paragraph we check that $\mathcal{G}$ is canonically isomorphic to $f_i^{-1}\mathcal{F}$. We strongly encourage the reader to skip this paragraph. To check this we have to show there is a bijection $\mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{G}, \mathcal{H}) = \mathrm{Mor}_{Sh(\mathcal{C}_i)}(\mathcal{F}, f_{i,*}\mathcal{H})$ functorial in the sheaf $\mathcal{H}$ on $\mathcal{C}$ where $f_{i,*} = u_i^p$. A map $\mathcal{G} \to \mathcal{H}$ is the same thing as a compatible system of maps

$$\varphi_{a,b,V_j} : f_{a\circ b}^{-1}\mathcal{F}(u_b(V_j)) \longrightarrow \mathcal{H}(u_j(V_j))$$

for all $a : j \to i$, $b : k \to j$ and $V_j \in \mathrm{Ob}(\mathcal{C}_j)$. The compatibilities force the maps φ_{a,b,V_j} to be equal to $\varphi_{a\circ b,\mathrm{id},u_b(V_j)}$. Given $a : j \to i$, the family of maps $\varphi_{a,\mathrm{id},V_j}$

corresponds to a map of sheaves $\varphi_a : f_a^{-1}\mathcal{F} \to f_{j,*}\mathcal{H}$. The compatibilities between the $\varphi_{a,\mathrm{id},u_a(V_i)}$ and the $\varphi_{\mathrm{id},\mathrm{id},V_i}$ implies that φ_a is the adjoint of the map φ_{id} via

$$\mathrm{Mor}_{Sh(\mathcal{C}_j)}(f_a^{-1}\mathcal{F}, f_{j,*}\mathcal{H}) = \mathrm{Mor}_{Sh(\mathcal{C}_i)}(\mathcal{F}, f_{a,*}f_{j,*}\mathcal{H}) = \mathrm{Mor}_{Sh(\mathcal{C}_i)}(\mathcal{F}, f_{i,*}\mathcal{H})$$

Thus finally we see that the whole system of maps φ_{a,b,V_j} is determined by the map $\varphi_{id} : \mathcal{F} \to f_{i,*}\mathcal{H}$. Conversely, given such a map $\psi : \mathcal{F} \to f_{i,*}\mathcal{H}$ we can read the argument just given backwards to construct the family of maps φ_{a,b,V_j}. This finishes the proof that $\mathcal{G} = f_i^{-1}\mathcal{F}$.

Assume (0.2.1) holds. Then the functor $\mathcal{F} \mapsto f_i^{-1}\mathcal{F}(U)$ commutes with finite limits because finite limits of sheaves are computed in the category of presheaves (Lemma 00W2), the functors f_a^{-1} commutes with finite limits, and filtered colimits commute with finite limits. To see that $\mathcal{F} \mapsto f_i^{-1}\mathcal{F}(V)$ commutes with finite limits for a general object V of $\mathcal{C}$, we can use the same argument using the formula for $f_i^{-1}\mathcal{F}(V) = \mathcal{G}(V)$ given above. Thus f_i^{-1} is left exact and the proof of the lemma is complete. □